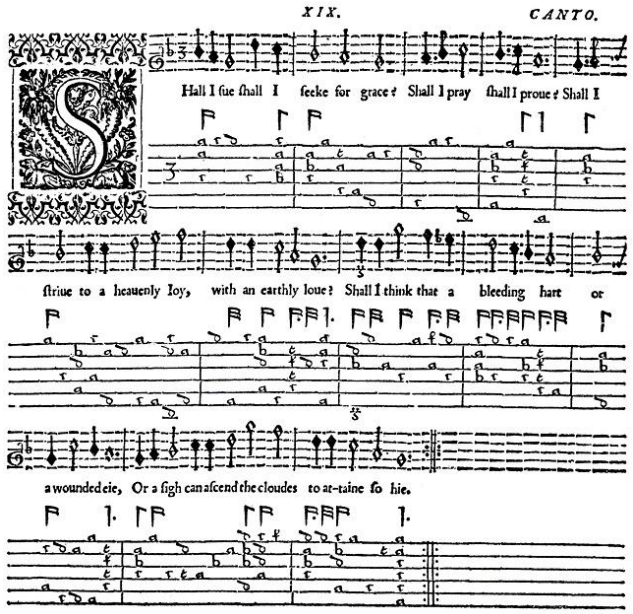
# MIR in Practice:
## The pre-histories of Music Information Retrieval

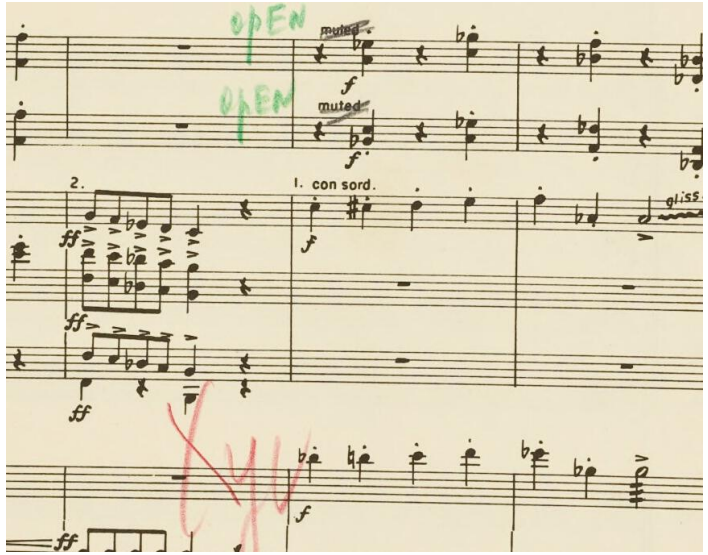
Eamonn Bell
Department of Computer Science
Durham University
eamonn.bell@durham.ac.uk

# What is MIR?

> Music information retrieval (MIR) is a research field that aims, among other things, at automatically extracting semantically meaningful information from various representations of music entities, such as a digital audio file, a band's Web page, a song's lyrics, or a tweet about a microblogger's current listening activity.

(Schedl, [c. 2012])




(Clayton et al. 2020)

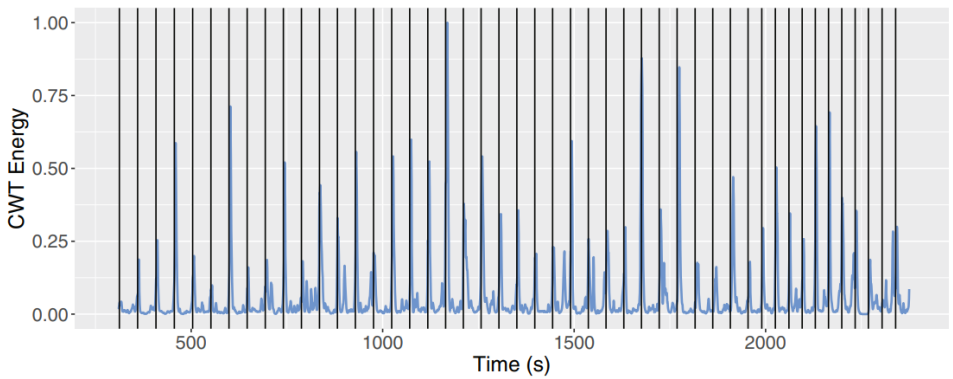



FIG. 10. Pairwise CWT Energy for the vocalist and tabla player in VK_Multani, khyal in vilambit ektal, from the NIR corpus, plotted over time (onsets of metrical cycles labeled via vertical lines).

[redacted] 6 months ago

I love the face of the timpani player at 53:46, he is so happy to play :)

5 REPLY

```
X: 1
T: The Tarbolton
R: reel
M: 4/4
L: 1/8
K: Edor
D|Eeed e2 BA|GBAF GFEF|Dddc d2 AF|GBAG FDDF|
Eeed efge|fedf edBA|GABG FGA=c|BGAF GE E:|
gfef (3gfe be|gebe gfef|d2 fd adfd|ABAG FDDF|
(3GGG BG (3FFF AF|Eeef gfed|B2 dB AGFA|BGAF GE E:|
```

Activity at time of memory (open response)

Please describe the **memory** associated with this music. Please give as much detail as possible (such as **who** you were with, **where** you were, and **what** you were doing in the remembered event) (open response)

(Jakubowski and Ghosh 2021)

# (Non-)History of MIR

| | |
|---|---|
| First ISMIR conference | 2000 |
| First MIREX | 2005 |

(Scherzinger 2019)

# (Non-)History of MIR

Kassler, Michael. “Toward Musical Information Retrieval.” *Perspectives of New Music* 4, no. 2 (1966): 59. https://doi.org/10.2307/832213.

# (Non-)History of MIR

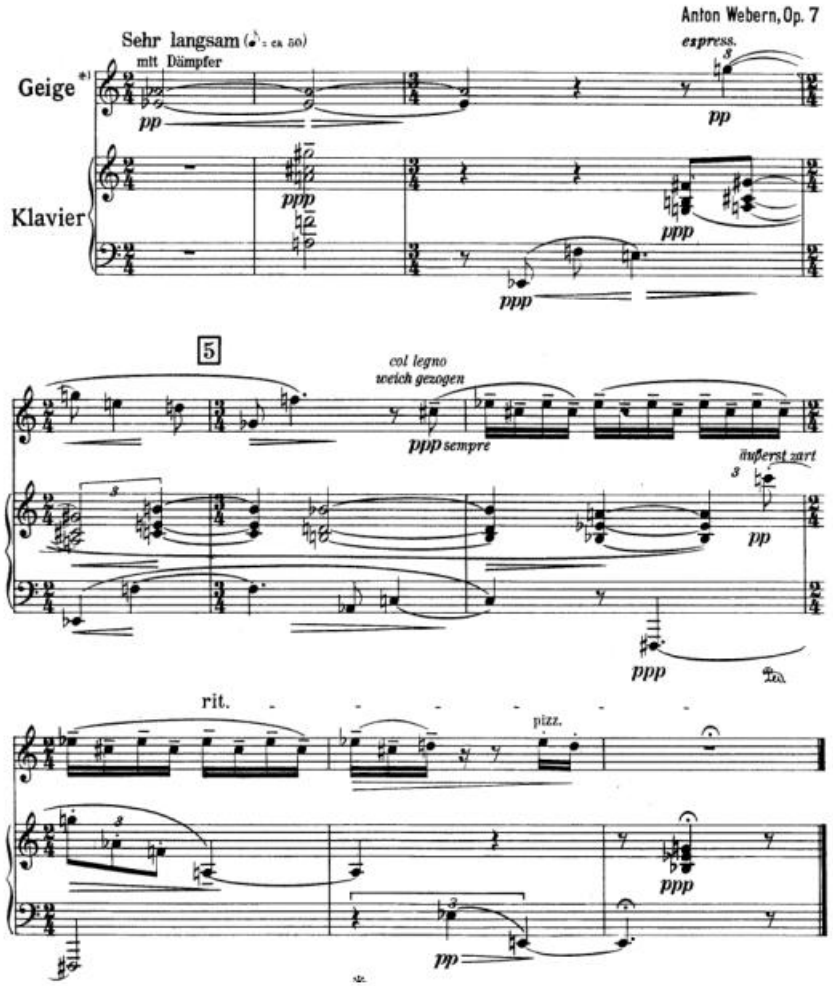
| | |
|---|---|
| [Fast Fourier Transform (Tukey/Cooley) | 1965] |
| **Kassler (coinage)** | **1966** |
| | |
| "computational musicology" | 1960s |
| transfer from speech research | 1970s |
| "a relative lull" | 1980s |
| | |
| query-by-humming, databases etc. | 1990s |
| Kageyama et al. (1993); McNab et al. (1996) | |
| | |
| **First ISMIR conference** | **2000** |
| First MIREX | 2005 |

(Burgoyne et al. 2015)

# Digital Alternate Representation of Musical Scores (DARMS)

## INPUT LANGUAGE CODE

```
WEBERN,FOUR PIECES FOR VIOLIN AND PIANO, OP. 7 (1910)
2.101*VIOLIN,/102*PIANO,/                                          EOF
101 *XG,*XT*$SEHR LANGSAM (E=CA 50)$ *XM2*.4 VPP*L1,21-HJ,24-N4J,*AMIT *
$DAEMPFER*A / V*L1*G2,21HJ,24N4J *G2 / *XM3*.4 21Q RQ *X3E1*.2 RE1,*XVES
PRESS.$ VPP,30**Q1JL+1 / *XM2*.4 102 *XG *XM2*.4 RW / VPPP,24**H,26**H,3
0*=H*- / *XM3*.4 RQ RQ (VPPP,16**E,18**E,22*=EL+1 17**EJ,19*=EJ,23*=EJ)
/ *XM2*.4 ** *XF *XM2*.4 RW / 79**H,84**H*- / *XM3*.4 RE VPPP*L1,69-EL+1
 77**E V*L1*G2,76**Q.L+1 V*G2 / *XM2*.4 101 *XG *XM2*.4 V*L1,30**E 29**Q
 V*L1 27**E / *XM3*.4 V*G1,23-E 29**Q.L+1 V*G1 RE,*XVWEICH GEZOGEN$,*XVC
OL LEGNO$ VPPPSEMPRE,26*=EL+1*- / ((28-S*- 26*=S*- 28S*- 26SL+1*-)) ((28
SL+1*- 26S*- 28S*- 26S*-)) ((28S*- 26S*- 28S*- 26S*-)) / *XM2*.4 102 *XG
 *XM2*.4 *X3Q2*.2 V*L2,17**H2,19*=H2,23*=H2 V*L2,19**Q2J,21**Q2J,25**Q2J
 / *XM3*.4 19Q,21Q,25Q 18**HJ,20**HJ,25-HJ / V*G2,18Q,20Q,25Q V*G2,18-QJ
,21-QJ,24**QJ 18Q1,21Q1,24Q1L+1 VPP,33**E1L+10,*XVAEUSZERST ZART$ / *XM2
*.4 ** *XF *XM2*.4 V*L2,69-QL+1 V*L2,77**QJ / *XM3*.4 77Q. V*G2 72-E 74*
*QJ V*G2 / 74L+1 RE VPPP,63*=Q.L+1 / *XM2*.4 101 *XG *XM2*.4 ((28-SL+1*-
 26*=S*- 28S*- 26S*-)) ((28S*-,*XTRIT.$ 26S*- 28S*- 26SL+1*-)) / ((V*G1,
28-SL+1*- 26*=S*- 27**SL+1*-)) RS RE ((28SQ V*G1,27SQ)) / RW // 102 *XG
*XM2*.4 (V*G2,30**E10 24-E10 21**E10) V*G2,17**QJL+1*- / 17Q RQ RE VPPP,
18-Q,21-Q,23**Q RE // ** *XF *XM2*.4 63*=H / RQ2 VPP*G2,76-Q2L+1 V*G2,19
**Q2JL+1 / 19Q. RE //                                              EOF
```




(Bauer-Mengelberg and Ferentz 1968; Selfridge-Field 1997)

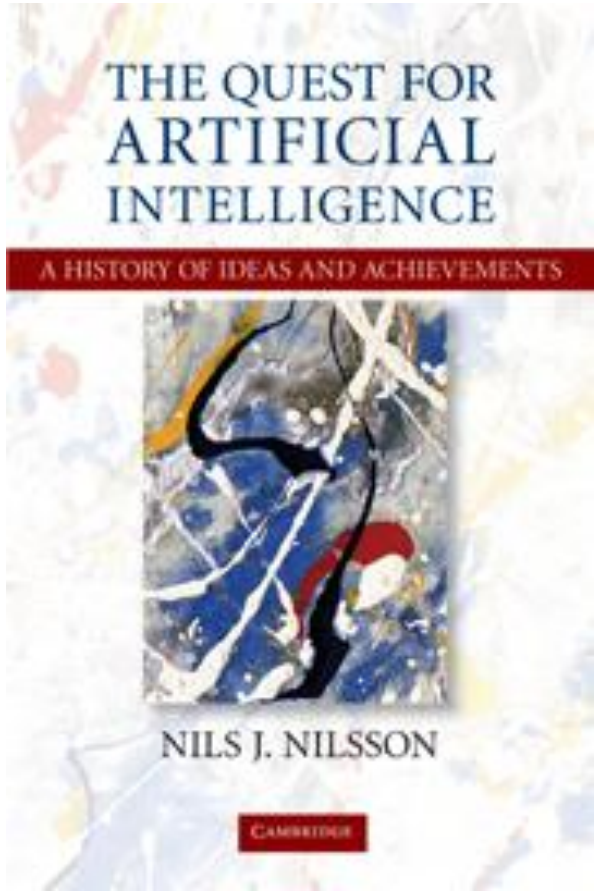
```
          CODE = ‘,/$+-$+-*-+,Y$()Z’
          KEY = ‘A*G)H/I$N(RYS+T,-.Z’
BACK      CODE *CH/’1’* = /F(PRINT)
          KEY  *TR/’1’* CH
          DECODED.CODE = DECODED.CODE TR /(BACK)
PRINT     SYSPOT = ‘ DECODED CODE IS - - - ‘
               DECODED.CODE
END
```

Allen Forte, “The Programming Language SNOBOL3: An Introduction.” *Computers and the Humanities* 1, no. 5 (1967), 160.

# What can history do?

- Explain social, cultural, economic change with reference to the past
- Identify "hidden" stories and actors
- Avoid repeating mistakes
- Find inspiration
- **Examine whether things could have gone another way**
  - Antidote to determinism of various kinds (technological, economic, etc.)
  - History of science/technology (technoscience) has to also deal with the role of nature here (Rheinberger 2010)

# SOTA for AI history?




(Nilsson 2009)

Ali, S.M. *et al.* (2023) 'Histories of artificial intelligence: a genealogy of power', *BJHS Themes*, 8, pp. 1–18.

# What can history do for ML/AI/ADPs?

Historians are experts in processing **archives** := "collections of materials, historical and current, systematically stored for academic, scholarly, heritage, and legacy purposes"

(Jo and Gebru 2020)

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Table 2 - Parallels in models of archiving and data collection (Jo and Gebru 2020)

| **Archives** | **ML Data Collection** |
|---|---|
| Collection Policy & Mission Statements | |
| Archival Consortia | |
| Participatory & Community Archives | |
| Codes of Conduct & Ethics | |
| Appraisal Guidelines & Records | |

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| **Archives** | **ML Data Collection** |
|---|---|
| Collection Policy & Mission Statements | Model Objectives, User-centered Needs |
| Archival Consortia | Data Trusts |
| Participatory & Community Archives | Crowdsourcing |
| Codes of Conduct & Ethics | AI Codes of Ethics |
| Appraisal Guidelines & Records | Datasheets for Datasets |

# Peek ahead

FATE/FAccT movement in AI recently concerned with defining new artifacts to accompany models and data (cf. "data provenance" movement in databases in the early 2000s)

e.g. Datasheets for Datasets, Model Cards for Model Reporting…

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FATE/FAccT movement in AI recently concerned with defining new artifacts to accompany models and data (cf. "data provenance" movement in databases in the early 2000s)

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…but the history of models and of datasets is conspicuously absent, not to mention that of machine learning as coherent field (Plasek 2016)

# Peek ahead

Though there is some interest in "archives" (Jo and Gebru 2020) and the "genealogy" of datasets (Denton et al. 2020)

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Though there is some interest in "archives" (Jo and Gebru 2020) and the "genealogy" of datasets (Denton et al. 2020)

…there is no concrete proposal for **how the history of ML/AI/ADP processes themselves** should be represented in the context of scientific publications describing functioning models/datasets

# Open areas of research

- What is the "computational attitude"?
- Solving the all-interval twelve-tone series problem
  - An interesting problem in combinatorics, still used today to benchmark CSP solvers
- Analysing music as information
  - Several case studies (c. 1948-1960), including Pierce/Shannon, Quastler, Fucks
  - **All early applications of information theory (Moles)**
- "Syntax-oriented" processing of musical scores
  - **Early experiments in musical score segmentation at MIT with SNOBOL (Forte)**

Currently (2019 onward)

- **Using music to help understand ill-structured problems in early AI in IPL-V** (Reitman and Sanchez)
- Early attempts to model Schenkerian analysis of tonal music (Kassler)

Photo: Yves Tessier
December 1972
Université Laval, QC

# Case A
# Abraham Moles: pre-digital MIR pioneer?

# MIR *avant la lettre*

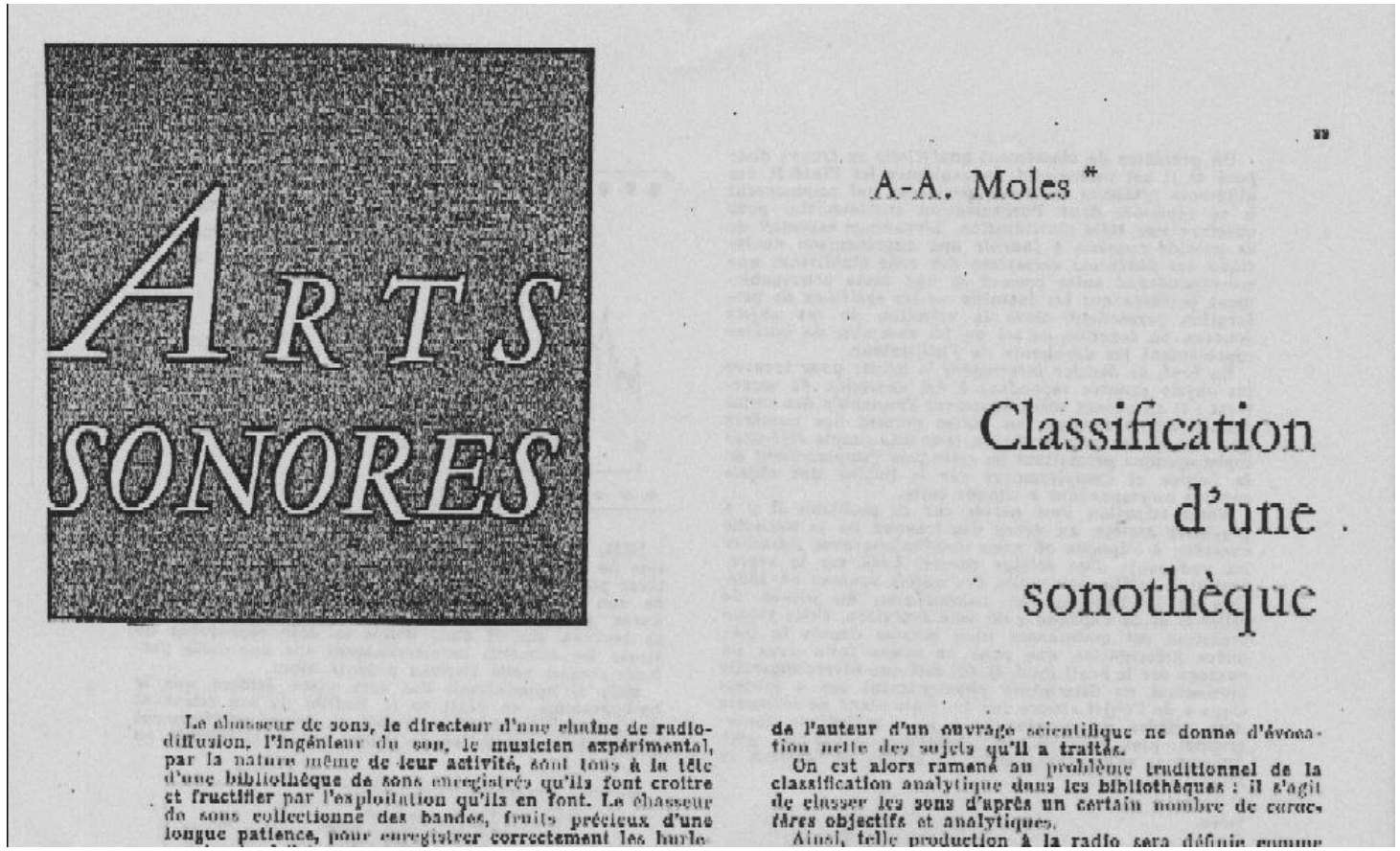
"I want a recording of *t* minutes, of theatre, spoken-word, including musical sequences, poetic [*de la poésie*], and using four actors (of which three [have] feminine voices), of an elevated cultural register, etc"

(Moles 1959)

Pierre Schaeffer (1910-1995)
"sampling"

shift from discs to tape
(Bohlman and McMurray 2017)

# Punch-card MIR



ARTS SONORES

A-A. Moles *

Classification d'une sonothèque

Le chasseur de sons, le directeur d'une chaîne de radio-diffusion, l'ingénieur du son, le musicien expérimental, par la nature même de leur activité, sont tous à la tête d'une bibliothèque de sons enregistrés qu'ils font croître et fructifier par l'exploitation qu'ils en font. Le chasseur de sons collectionne des bandes, fruits précieux d'une longue patience, pour enregistrer correctement les hurle-

de l'auteur d'un ouvrage scientifique ne donne d'évocation nette des sujets qu'il a traités.

On est alors ramené au problème traditionnel de la classification analytique dans les bibliothèques : il s'agit de classer les sons d'après un certain nombre de caractères objectifs et analytiques.

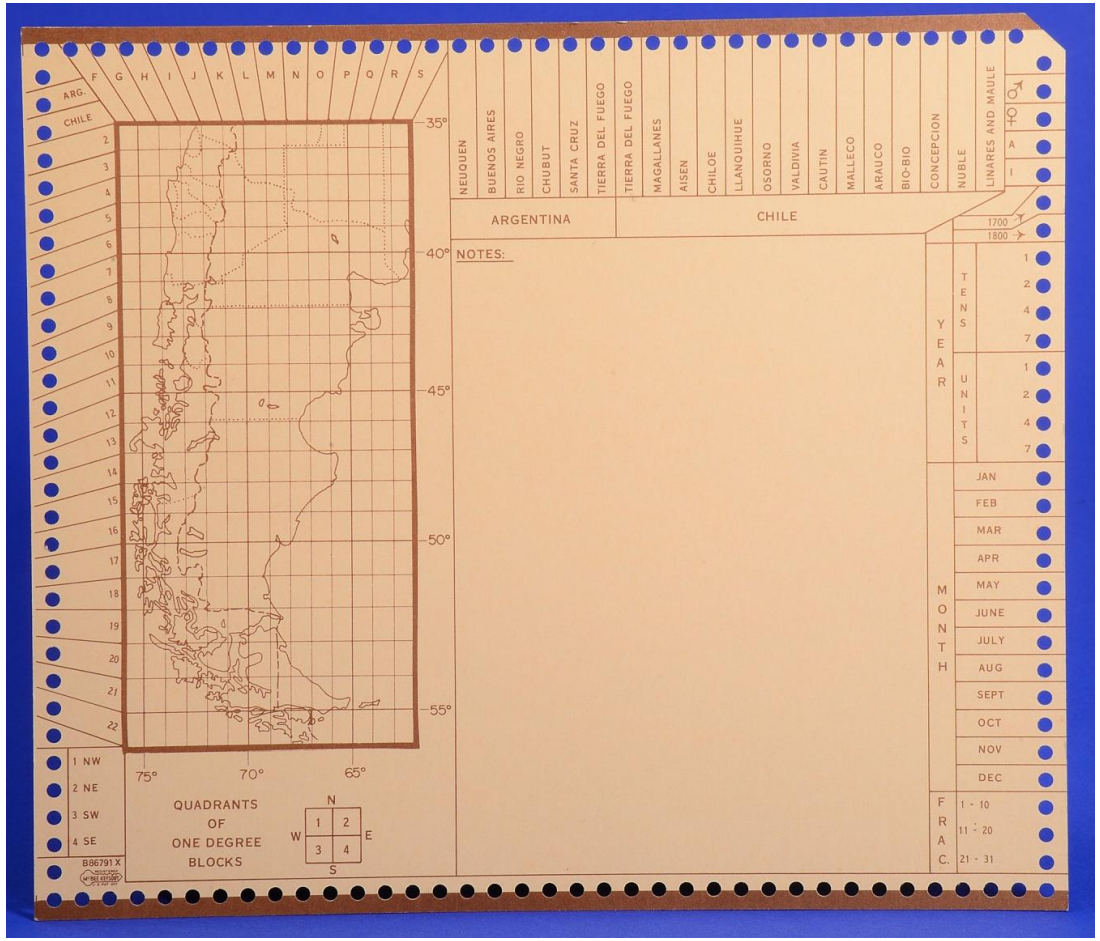
Ainsi, telle production à la radio sera définie comme

**Pierre Schaeffer (1910-1995)**
"sampling"

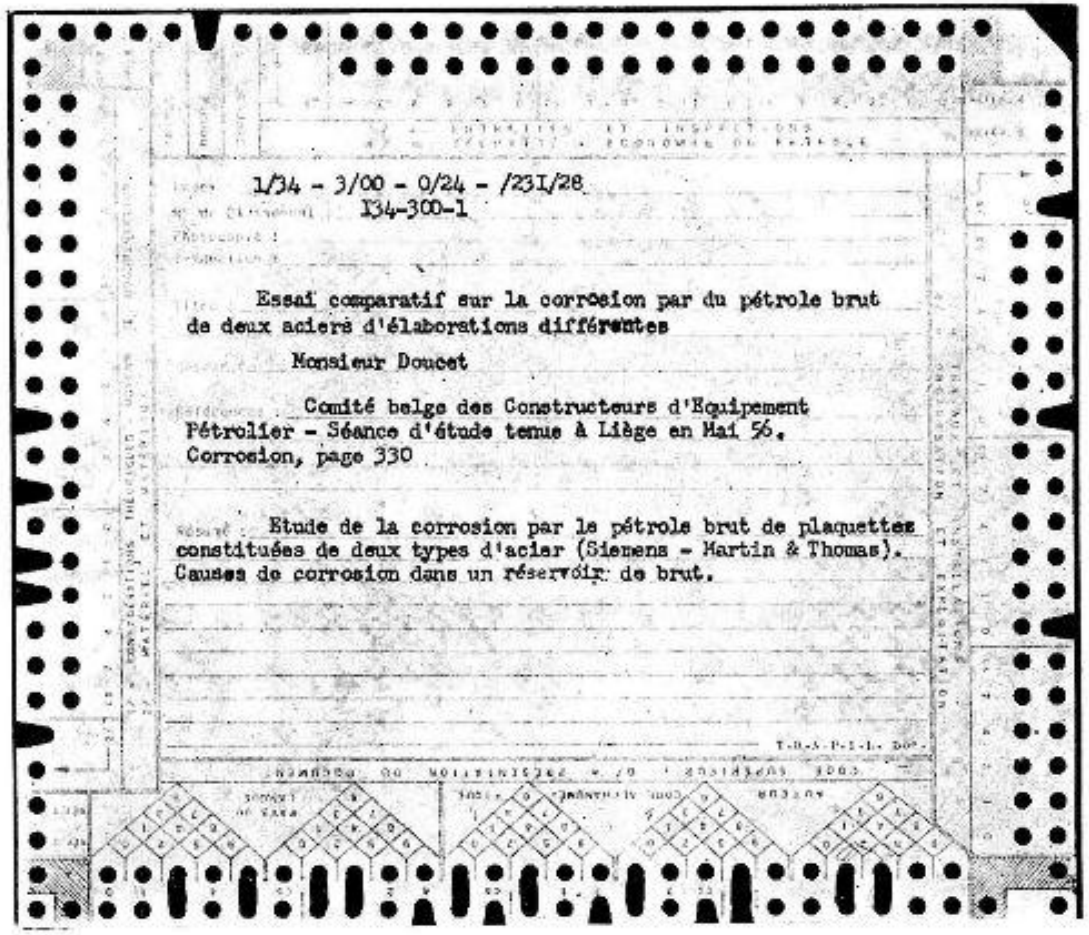
shift from discs to tape
(Bohlman and McMurray 2017)

Moles, Abraham. "Classification d'une sonothèque par cartes perforées" *Revue du Son*, no. 73 (May 1959): 136–39.

# Punch-card MIR

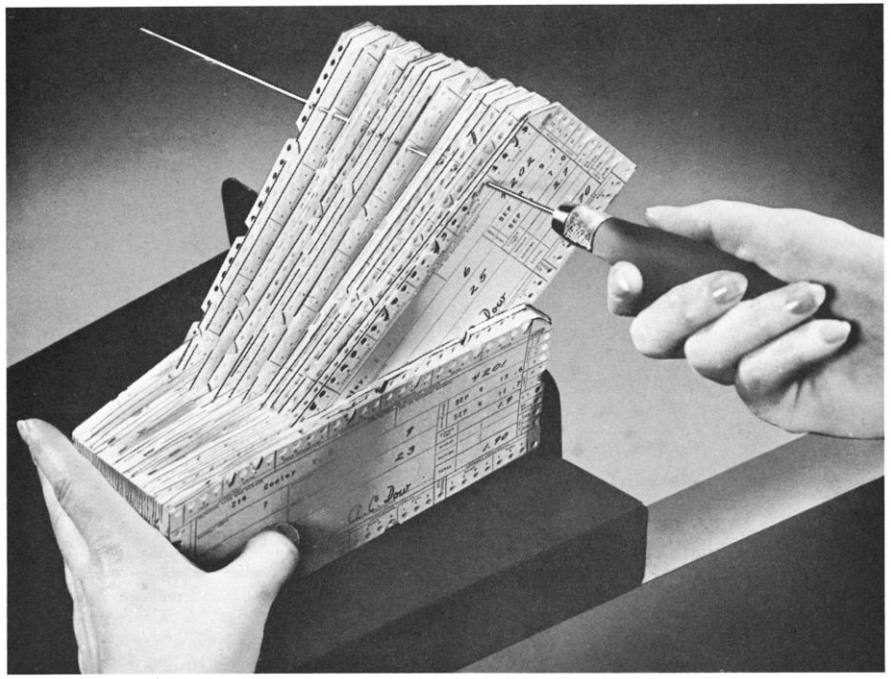



McBee Keysort Punch Card (c. 1960)
National Museum of American History
[IIIF Manifest]

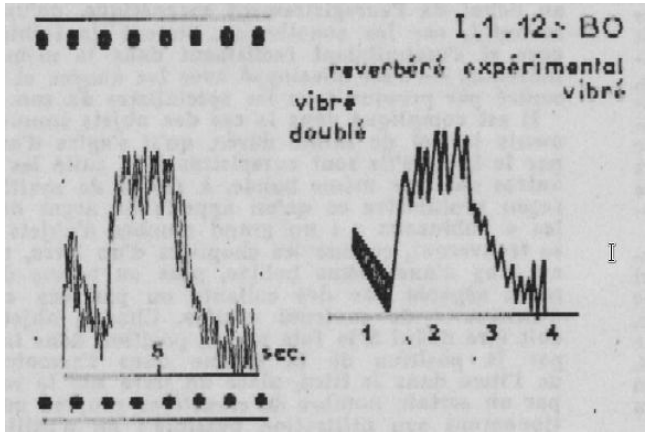


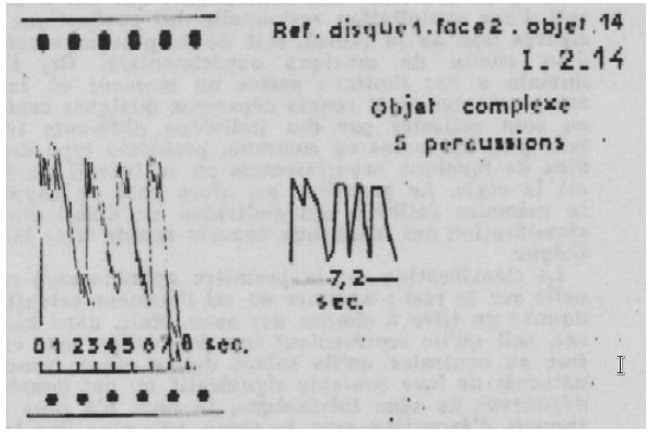

*Mécanalyse* card (Keysort?)
(Poindron and Salvan 1957)

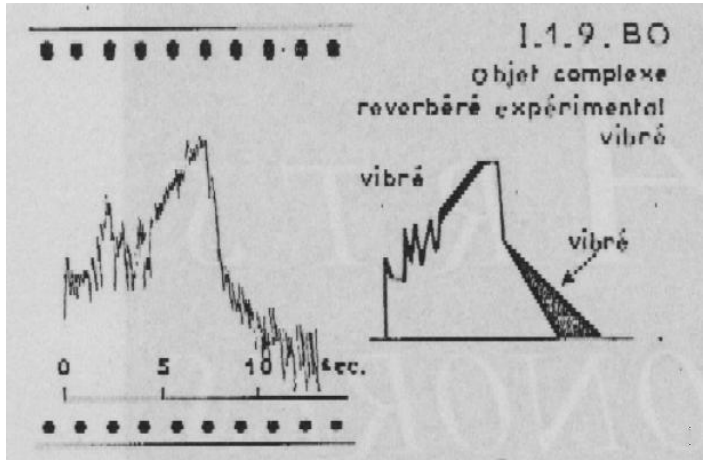
# Punch-card MIR



(Bourne 1963)



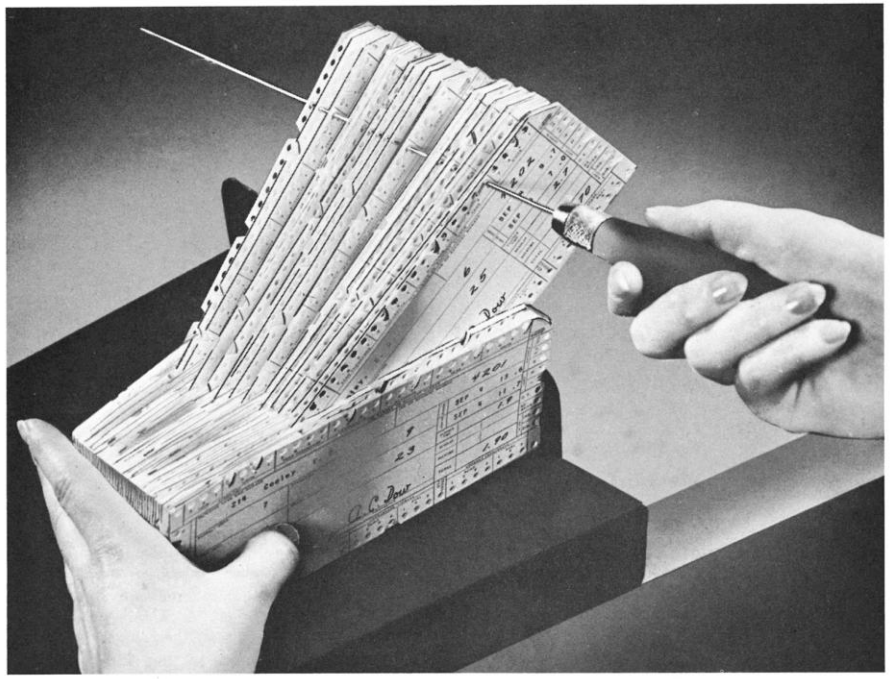




(Moles 1959)

**Edge-Notched Cards: A queryable database…made of paper (Soren Bjornstad, 2021)**
https://www.youtube.com/watch?v=v8qHPXPnQps

# Punch-card MIR



(Bourne 1963)

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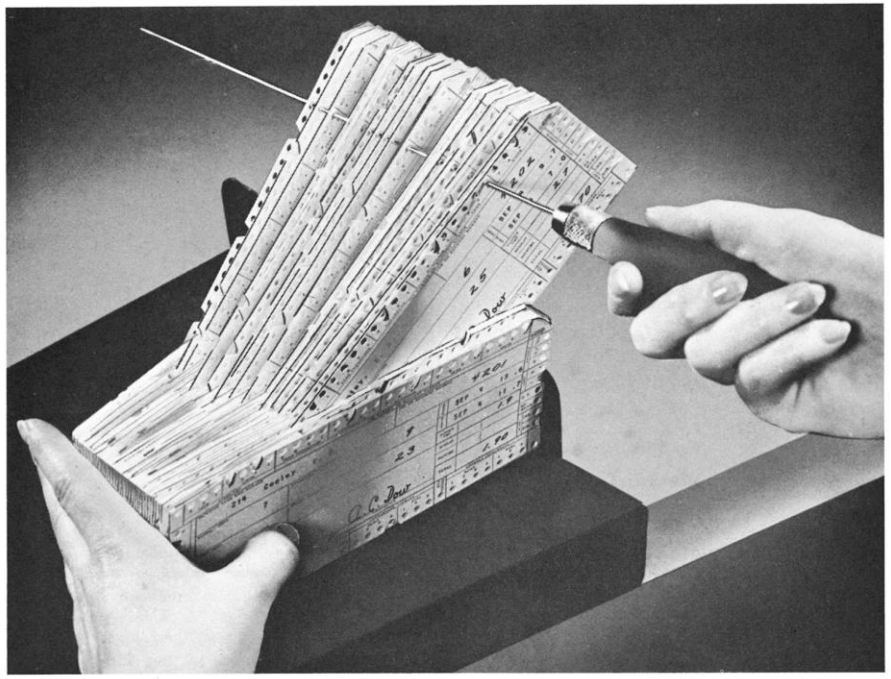

(Bourne 1963)

"I want an *objet sonore* longer than 8 seconds, with a single attack, medium dominant tonality, with 2 components comprising 1 percussion, amenable to reverb, with a thick timbre, comprising additionally a large number of harmonics, suggesting the sound of water, etc."

(Moles 1959)

# Cognitive cartoons

Moles, Abraham A., and Bernard Vallancien, eds. “Aspect temporel de l’intégration du message linguistique et modeles de l’audition.” In *Communications et langages*. Paris: Gauthier-Villars, 1963.

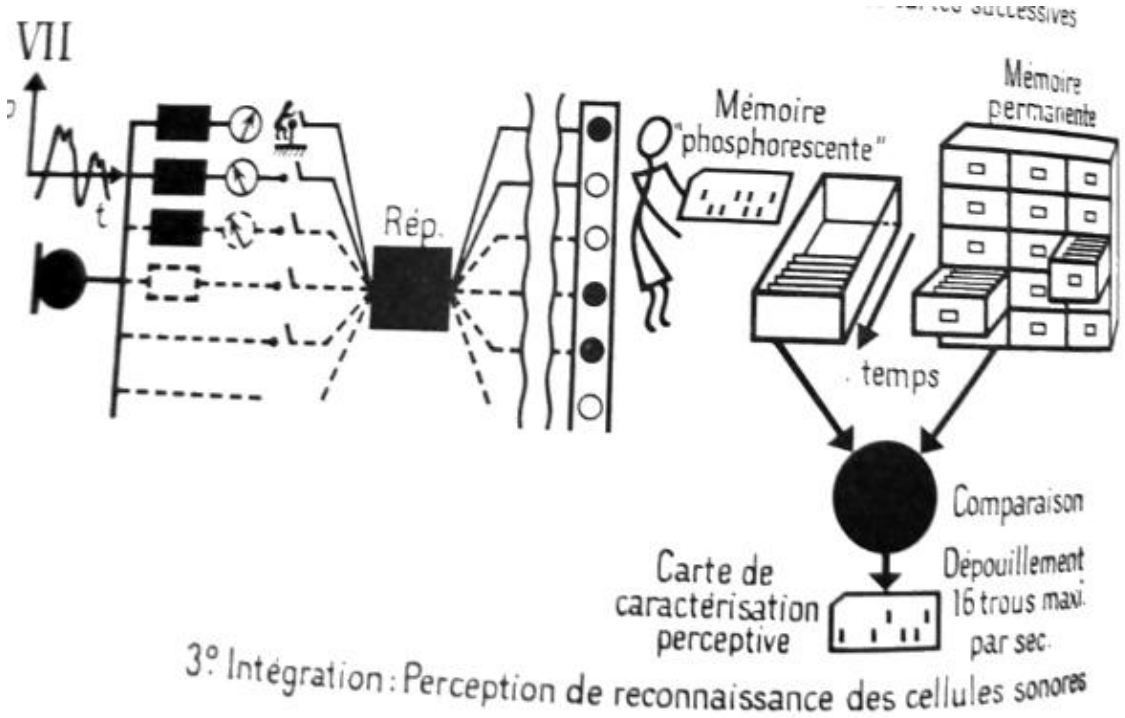


Fig. 5.

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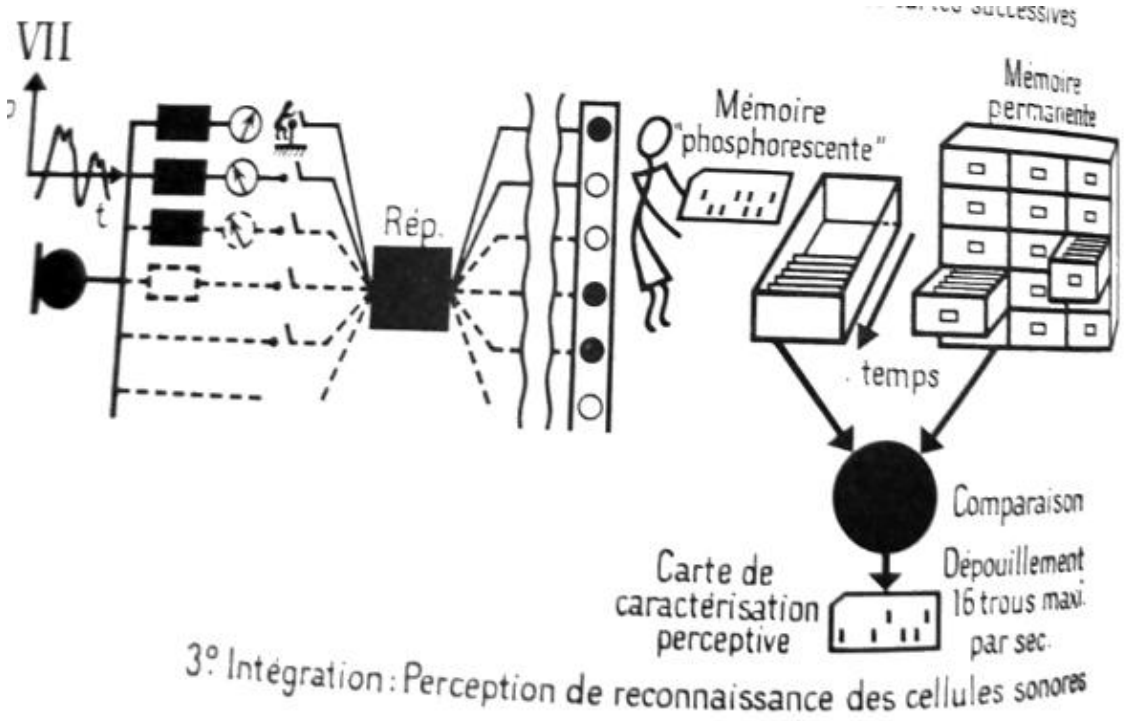
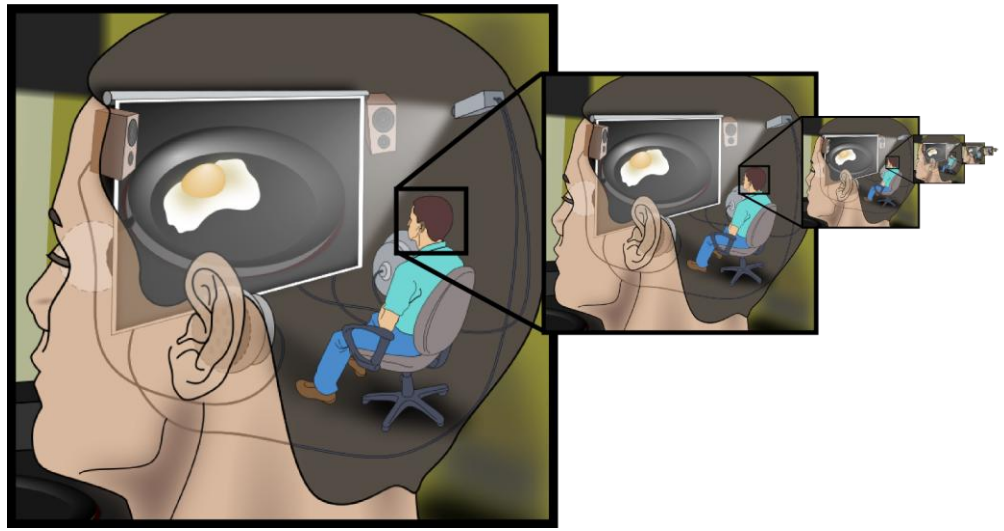



Fig. 5.



Original work: Jennifer Garcia (User:Reverie)Derivative work: User:Pbroks13Derivative work of derivative work: User:Was a bee, CC BY-SA 2.5 <https://creativecommons.org/licenses/by-sa/2.5>, via Wikimedia Commons

# Studies of concert programming

DATA

TABLE I: RELATIVE FREQUENCIES OF OCCURRENCE OF COMPOSER'S WORKS IN MUSIC PERFORMANCES

| No. | Composer | $p_c$ | No. | Composer | $p_c$ |
|---|---|---|---|---|---|
| | | *per cent* | | | *per cent* |
| 1 | Mozart | 6.1 | 40 | Couperin | 0.65 |
| 2 | Beethoven | 5.9 | 41 | Mahler | 0.6 |
| 3 | Bach | 5.9 | 42 | Rameau | 0.6 |
| 4 | Wagner | 4.2 | 43 | St. Saens | 0.6 |
| 5 | Brahms | 4.1 | | | |
| 6 | Schubert | 3.6 | | | |
| 7 | Handel | 2.8 | | | |
| 8 | Tchaikovsky | 2.8 | | | |
| 9 | Verdi | 2.5 | | | |
| 10 | Haydn | 2.3 | | | |
| 11 | Schumann | 2.1 | | | |

TABLE II. APPROXIMATE PROBABILITIES OF OCCURRENCE FOR VARIOUS WORKS OF SOME GREAT COMPOSERS

BEETHOVEN

| Works | $p_w$ | Works | $p_w$ |
|---|---|---|---|
| | *per cent* | | *per cent* |
| Symphony No. 5 | 4.2 | Symphony No. 2 | 1.5 |
| Symphony No. 7 | 3.1 | Coriolan Overture | 1.5 |
| Sonata No. 14 ("Moonlight") | 3.0 | Piano Concerto No. 3 | 1.5 |
| Symphony No. 8 | 2.7 | Piano Concerto No. 4 | 1.4 |
| Piano Concerto No. 5 | | Leonore Overture No. 3 | 1.4 |

(Moles 1966 [1958])

# Studies of concert programming

**Programme A**, low *H*

1. Beethoven,
   *Coriolan* Overture
2. Haydn,
   Military Symphony
3. Mozart,
   Jupiter Symphony
4. Beethoven,
   Piano Concerto No. 3
5. Beethoven,
   Symphony No. 5

**Programme B**, high *H*

1. Berg,
   Concerto for Violin and Orchestra
2. Schönberg,
   *Survivor from Warsaw*
3. Vivaldi,
   Quintet in F major
4. Busoni,
   Concerto for Violin, Op. 35
5. Britten,
   *Variations on a Theme of Frank Bridge*

# Studies of concert programming

"You are asked to underline the name of the one composer of each pair whose music you prefer in general, not taking the personality or greatness of the composer into consideration."

EXPERIMENTAL PROPORTIONS PHIL

| | Bach | Beethoven | Berlioz | Brahms | Chopin | Debussy | MacDowell | Cesar Franck | Grieg | Haydn |
|---|---|---|---|---|---|---|---|---|---|---|
| Bach | — | .453 | .074 | .432 | .063 | .179 | .032 | .053 | .032 | .12 |
| Beethoven | .547 | — | .084 | .432 | .074 | .168 | .021 | .063 | .042 | .10 |
| Berlioz | .926 | .916 | — | .968 | .484 | .895 | .126 | .547 | .274 | .82 |
| Brahms | .568 | .568 | .032 | — | .063 | .137 | .032 | .042 | .011 | .21 |
| Chopin | .937 | .926 | .516 | .937 | — | .726 | .242 | .568 | .284 | .80 |
| Debussy | .821 | .832 | .105 | .863 | .274 | — | .053 | .053 | .063 | .55 |
| MacDowell | .968 | .979 | .874 | .968 | .758 | .947 | — | .916 | .800 | .91 |
| Cesar Franck | .947 | .937 | .453 | .958 | .432 | .947 | .084 | — | .242 | .84 |
| Grieg | .968 | .958 | .726 | .989 | .716 | .937 | .200 | .758 | — | .89 |
| Haydn | .874 | .895 | .179 | .789 | .200 | .442 | .084 | .158 | .105 | — |
| Vic. Herbert | .979 | .968 | .842 | .979 | .905 | .947 | .621 | .874 | .811 | .94 |
| Mendelssohn | .947 | .947 | .347 | .947 | .411 | .705 | .084 | .432 | .147 | .82 |
| Mozart | .611 | .726 | .053 | .642 | .074 | .189 | .011 | .084 | .032 | .14 |
| Schubert | .821 | .895 | .116 | .853 | .105 | .358 | .000 | .116 | .011 | .45 |
| Schumann | .863 | .937 | .274 | .926 | .205 | .652 | .042 | .253 | .105 | .72 |

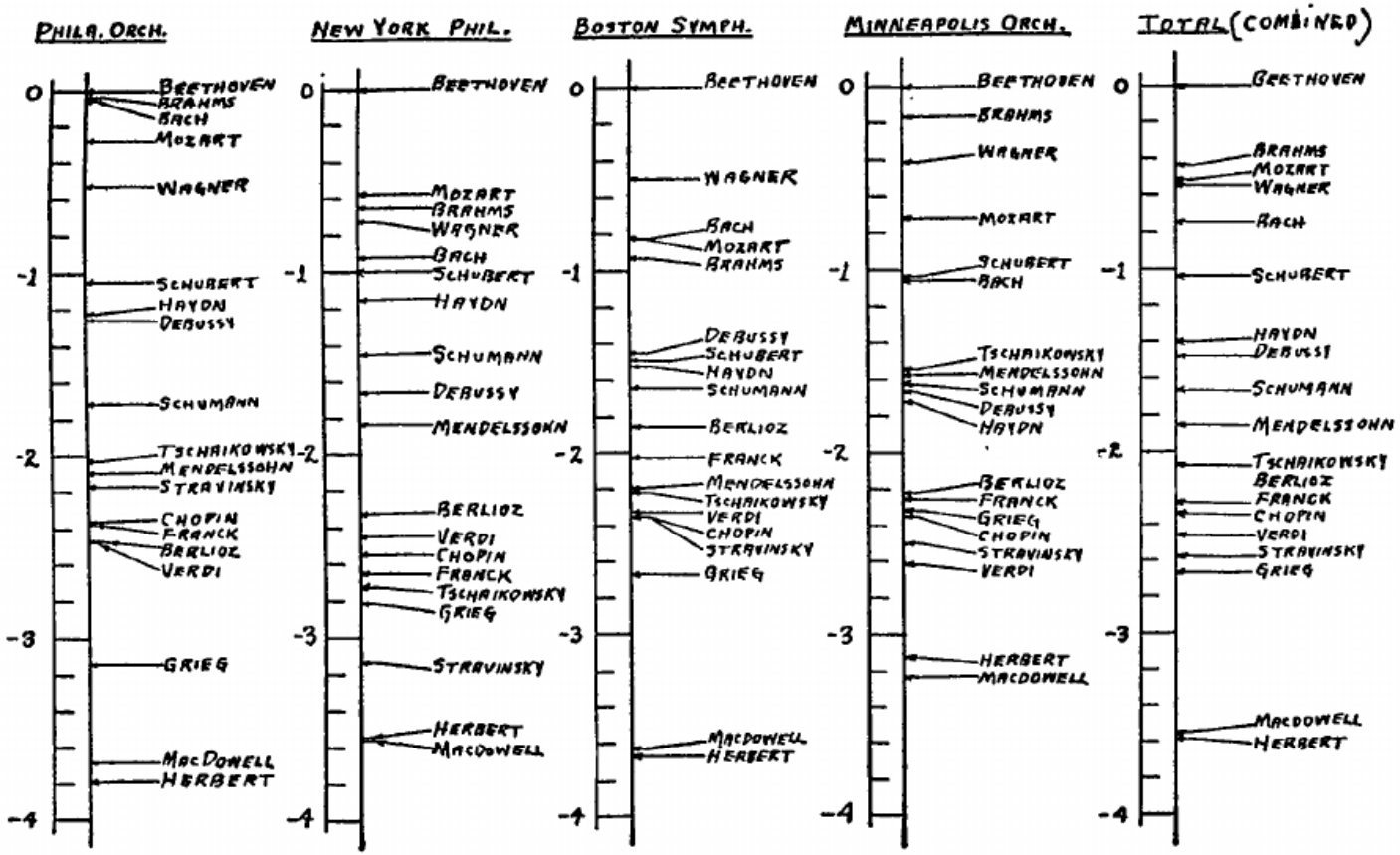
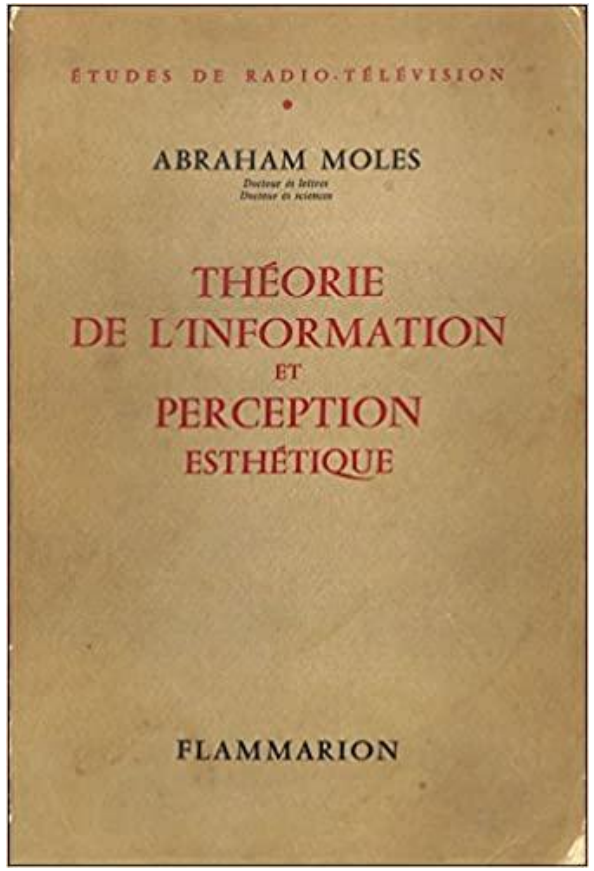


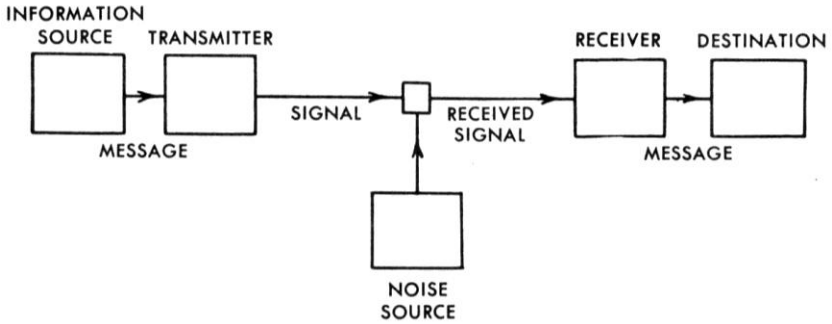

FIG. 1. Scales of Composer-Preferences.

**DATA**

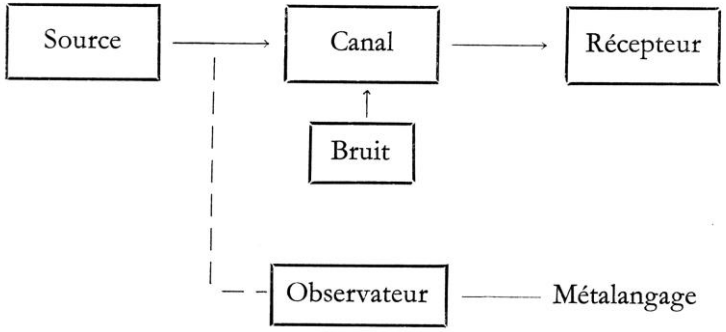
(Folgmann 1933)

# Information theory of (musical) listening




(Shannon 1948)




(Moles 1958, after Cherry 1957 [?])

Eamonn Bell, "Cybernetics, Listening, and Sound-Studio Phenomenotechnique in Abraham Moles's Théorie de l'information et Perception Esthétique (1958)," *Resonance* 2, no. 4 (December 1, 2021): 523–58, https://doi.org/10.1525/res.2021.2.4.523.

# Information theory of (musical) listening

> "we shall say that information or originality is a function of the improbability of the received message. **Considering this improbability as the received physical excitation**, we are led to connect information *H* to improbability *I* by Fechner's relation"

$$S = c \cdot \log R$$

"Weber-Fechner"

S : sensation intensity
R : stimulus intensity

(Moles 1966 [1958])

# Information theory of (musical) listening

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$$S = c \cdot \log R \quad \longrightarrow \quad H = -\log_2 \omega$$

"Weber-Fechner"

S : sensation intensity
R : stimulus intensity

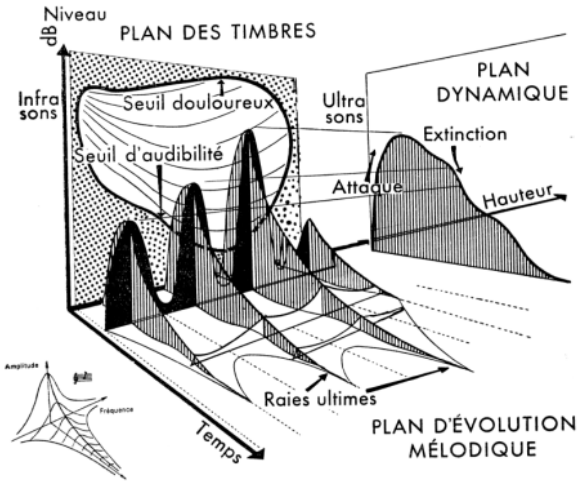
Shannon-Weaver

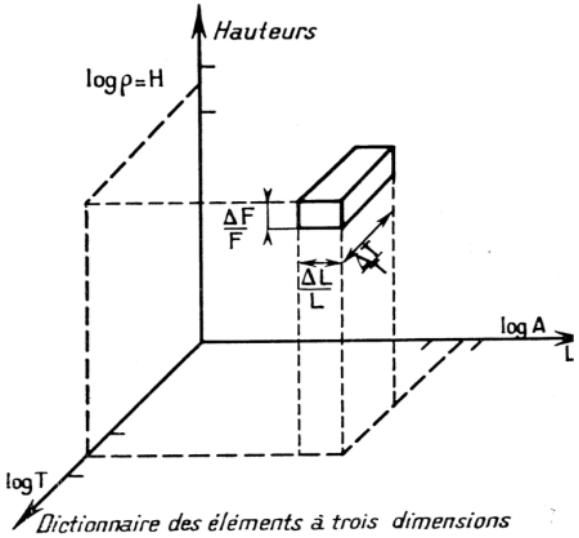
H : information q'ty of some event
ω : probability of some event

(Moles 1966 [1958])

(Relatedly, see, Mills 2011; Pias 2007)

# Information theory of (musical) listening

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(Relatedly, see, Mills 2011; Pias 2007)

# Case B
Walter Reitman and Marta Sánchez: Exchange between marginal AI and music composition in the 1960s

CIP Working Paper No. 37
November, 1960
Rough Draft

The Composition of a Fugue:
Protocol and Comments*

M. Sanches and W. R. Reitman

Sanchez, M., and W. R. Reitman. “The Composition of a Fugue: Protocol and Comments.” CIP Working Paper No. 37 (Rough Draft), November 1960. 157pp. Herbert A. Simon Collection, Carnegie Mellon University Library. Box 8. Folder 446.

- Walter Reitman
- b. 1932
- Michigan (PhD)
- **Asst. Prof.** Carnegie Tech (1957–1964?)
- Michigan MHRI
  → RPI afterward
- First taxonomy of
  "ill-defined problems"
  (Reitman 1964)

Credit: IRE Human Factors (1961)

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[WDP*] McCarthy (1956) →
[WDP] Minsky (1960) →
**[IDP] Reitman (1964)** →
[ISP] Simon (1973)

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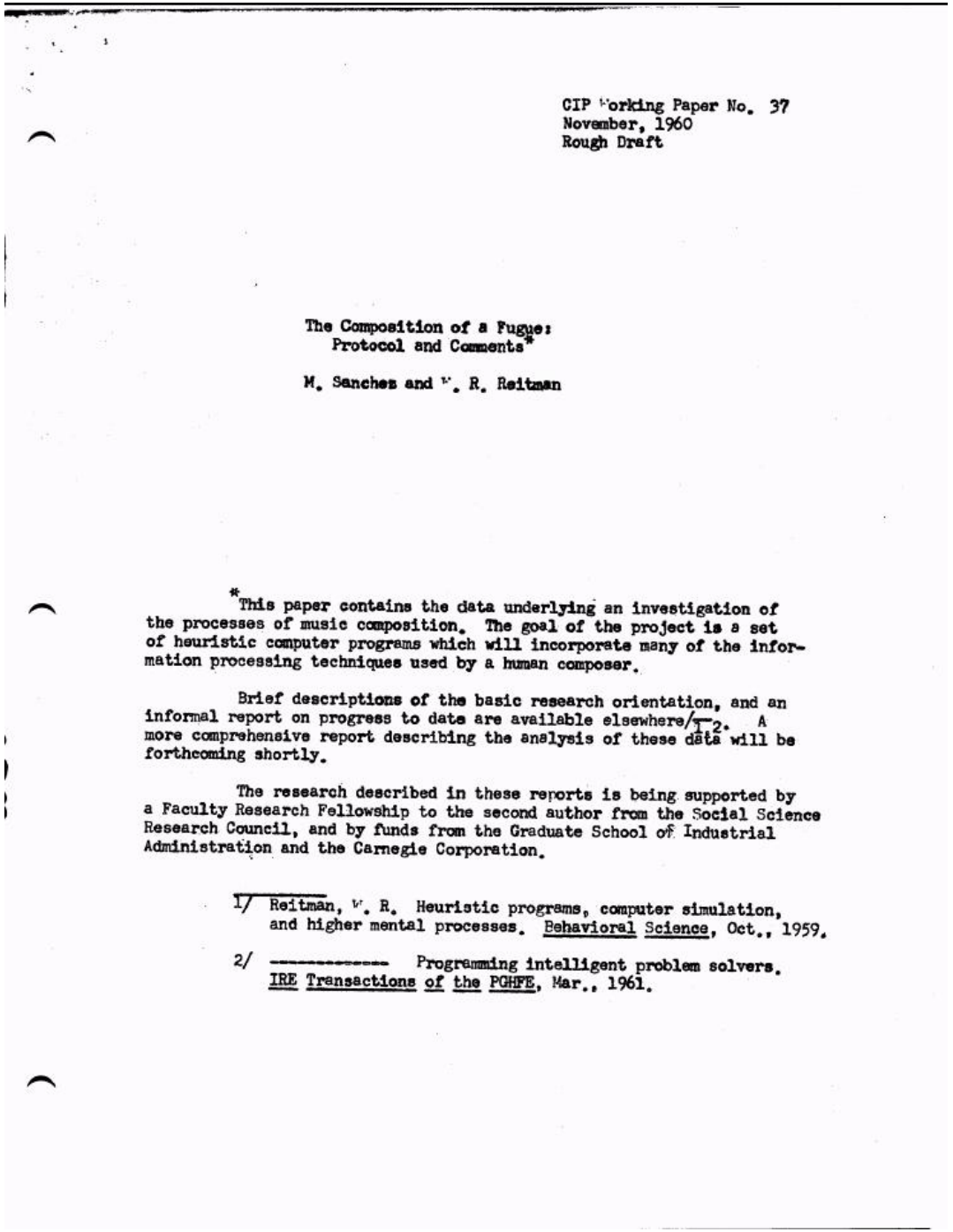
[WDP*] McCarthy (1956) →
[WDP] Minsky (1960) →
**[IDP] Reitman (1964)** →
[ISP] Simon (1973)

Credit: IRE Human Factors (1961)

- Marta Sánchez
- b. 1923
- d. 2006
- **Dalcroze instructor** (later Prof.) Carnegie Tech (1957–1997)

Credit: CMU



CIP Working Paper No. 37
November, 1960
Rough Draft

The Composition of a Fugue:
Protocol and Comments*

M. Sanches and W. R. Reitman

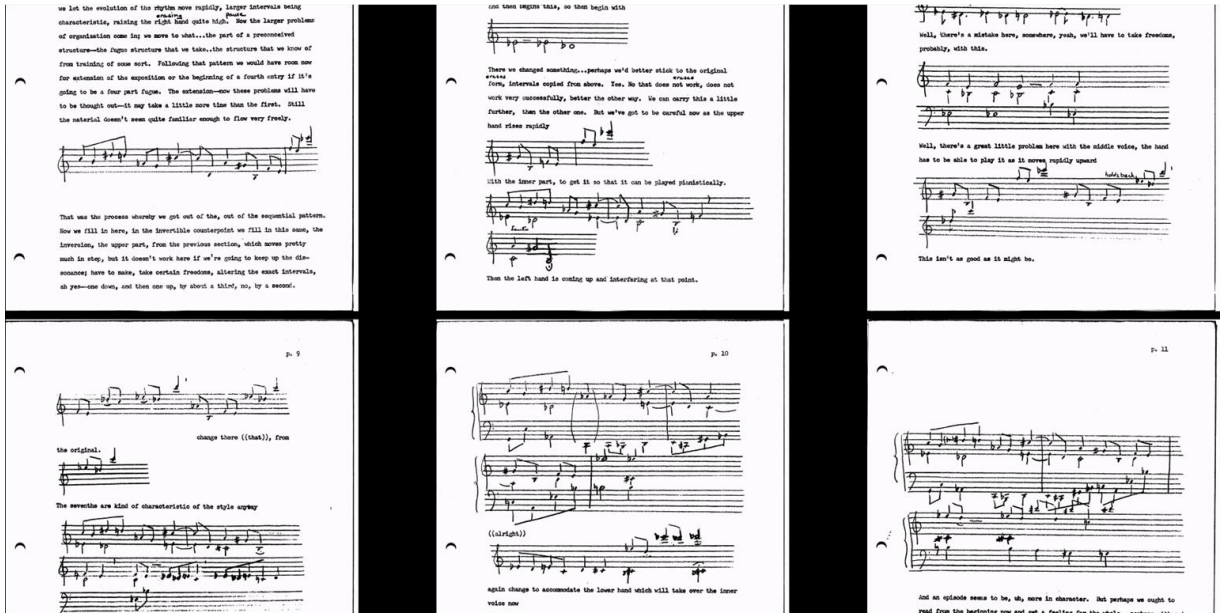
*This paper contains the data underlying an investigation of the processes of music composition. The goal of the project is a set of heuristic computer programs which will incorporate many of the information processing techniques used by a human composer.

Brief descriptions of the basic research orientation, and an informal report on progress to date are available elsewhere/1/2. A more comprehensive report describing the analysis of these data will be forthcoming shortly.

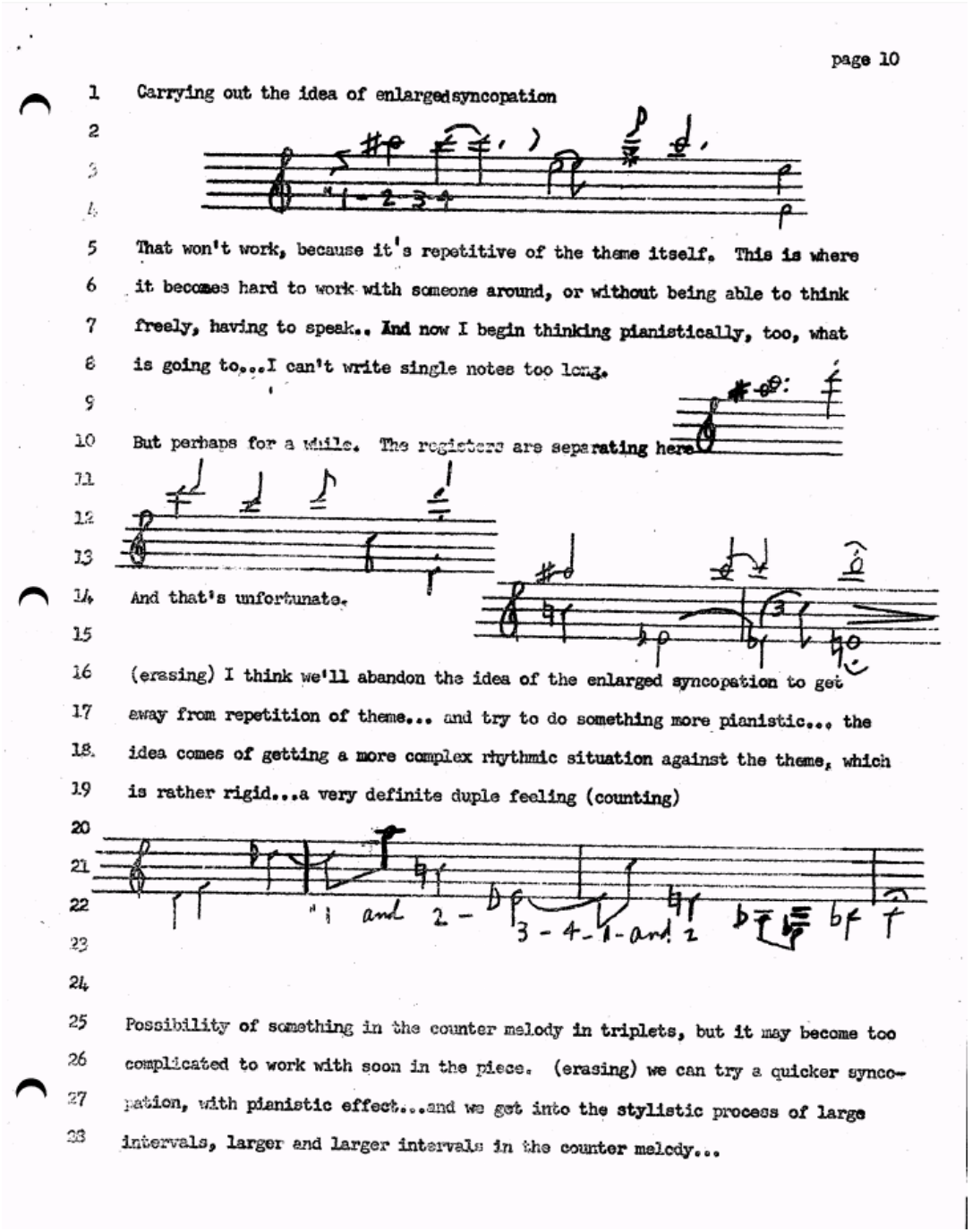
The research described in these reports is being supported by a Faculty Research Fellowship to the second author from the Social Science Research Council, and by funds from the Graduate School of Industrial Administration and the Carnegie Corporation.

1/ Reitman, W. R. Heuristic programs, computer simulation, and higher mental processes. Behavioral Science, Oct., 1959.

2/ ———— Programming intelligent problem solvers. IRE Transactions of the PGHFE, Mar., 1961.



Sanchez, M., and W. R. Reitman. “The Composition of a Fugue: Protocol and Comments.” CIP Working Paper No. 37 (Rough Draft), November 1960. 157pp. Herbert A. Simon Collection, Carnegie Mellon University Library. Box 8. Folder 446.



page 10

Carrying out the idea of enlargedsyncopation

That won't work, because it's repetitive of the theme itself. This is where it becomes hard to work with someone around, or without being able to think freely, having to speak.. And now I begin thinking pianistically, too, what is going to...I can't write single notes too long.

But perhaps for a while. The registers are separating here

And that's unfortunate.

(erasing) I think we'll abandon the idea of the enlarged syncopation to get away from repetition of theme... and try to do something more pianistic... the idea comes of getting a more complex rhythmic situation against the theme, which is rather rigid...a very definite duple feeling (counting)

Possibility of something in the counter melody in triplets, but it may become too complicated to work with soon in the piece. (erasing) we can try a quicker syncopation, with pianistic effect...and we get into the stylistic process of large intervals, larger and larger intervals in the counter melody...

Sanchez, M., and W. R. Reitman. "The Composition of a Fugue", 10.

Reitman, Walter R. *Cognition and Thought: An Information-Processing Approach*. New York: Wiley, 1965.

“Creative Problem Solving: Notes from an Autobiography of a Fugue”

> Our original aim was to see whether and how [fugue composition] might be brought within the framework of the General Problem Solver.
>
> **Our surrogate question was: How might we reproduce and account for the composer's behavior using GPS as a model of the thinking processes involved?** In other words, we used GPS as a tool for studying the behavior we observed in the protocol.
>
> **We found that there appeared to be important aspects of the behavior we could not encompass within the framework of the early GPS versions with which we worked**. The result was the evolution of our own framework [(Argus)], in many ways derived from the GPS model, but differing from it in some respects. In particular, the effort gave rise to the problem framework presented in Chapter 5 [“Problems”] and the active memory model of problem solving to be presented in Chapter 8 [“Argus: An Information-Processing Model of Thought”].
>
> (Reitman 1965, 180)

Reitman, Walter R., Richard B. Grove, and Richard G. Shoup. “Argus: An Information-Processing Model of Thinking.” *Behavioral Science* 9, no. 3 (1964): 270–81. https://doi.org/10.1002/bs.3830090312.

# GPS vs. Argus

**General Problem Solver**
(Newell and Simon 1961)

- "general", but focused on cryptarithmetic problems
- solves problems serially
- "single-minded"
- symbolic
- led to SOAR (Laird, Newell and Rosenbloom)

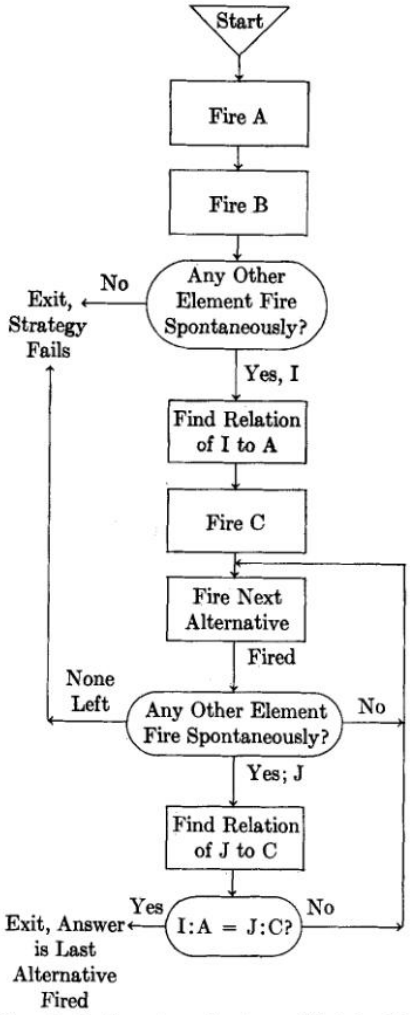



FIG. 1. Rudimentary Strategy (Slightly Simplified) for Analogy Problems of the Form *A:B :: C:* (*W*, *X*, *Y*, or *Z*). When an element is fired, activation and inhibition increments are added to related elements, altering their states.

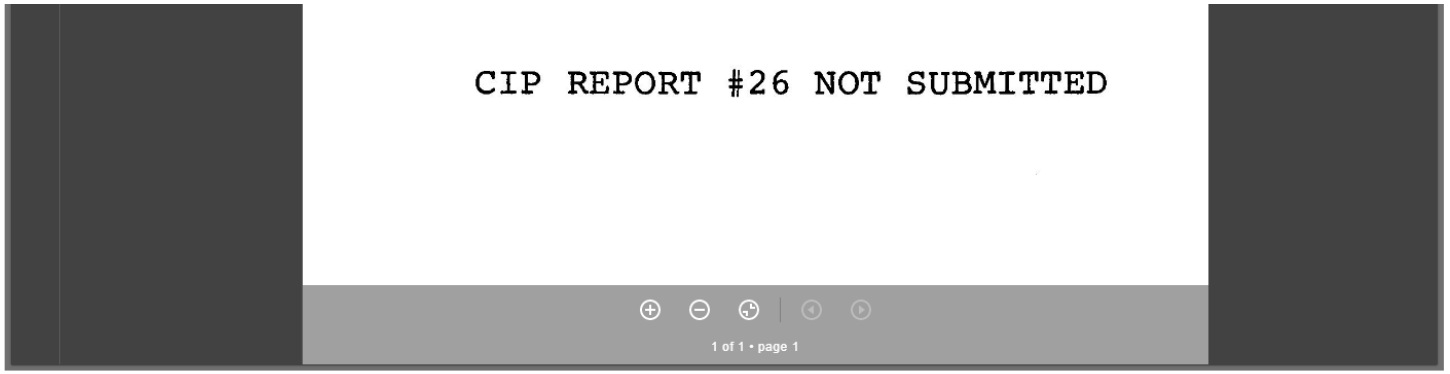
(Reitman et al. 1964)

**Argus**

(Reitman et al. 1964)

- solves analogies (A:B::C:__)
  - precursor to semantic networks?
- very simplistic parallel processing
- distractable, interruptable, forgetful
- neurosymbolic (?)
  - CAs have activation, inhibition, threshold, size, and **timestamp**
- led to…?

# Historical Source(s) for Argus

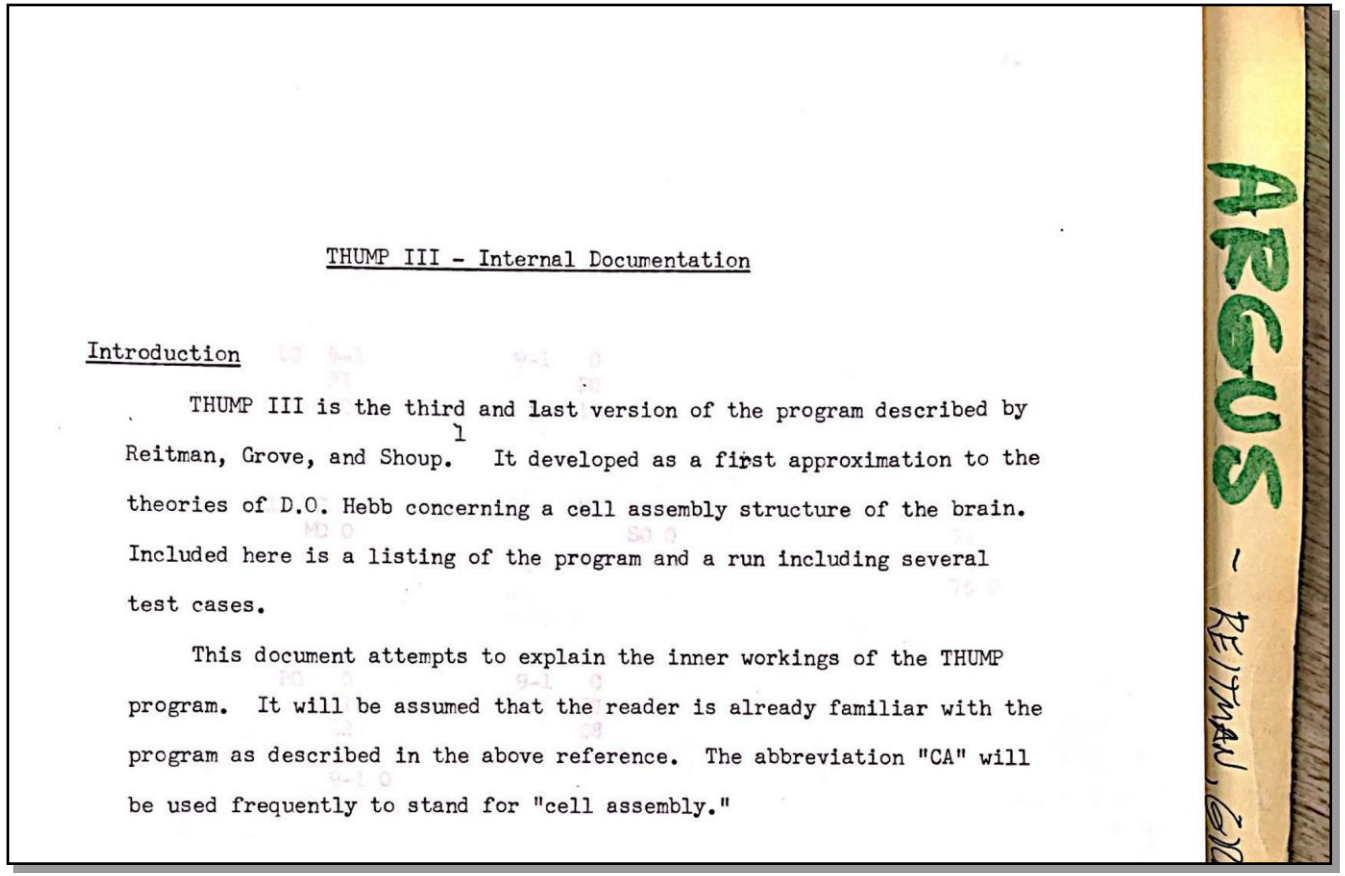



**Complex Information Processing (CIP) -- Paper #26 -- "The Processes of Musical Composition: A Preliminary Study" -- Reitman and Sanchez - administrative records**

COLLECTION
Herbert Simon

SERIES
Carnegie-Mellon University

# Historical Source(s) for Argus

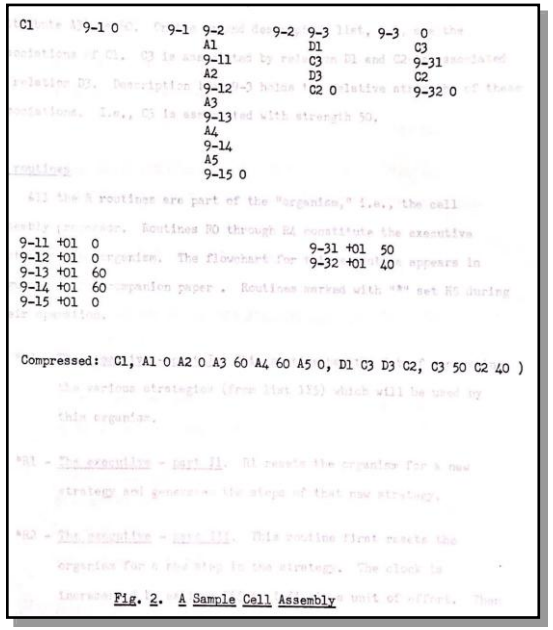


THUMP III - Internal Documentation

Introduction

THUMP III is the third and last version of the program described by Reitman, Grove, and Shoup.[1] It developed as a first approximation to the theories of D.O. Hebb concerning a cell assembly structure of the brain. Included here is a listing of the program and a run including several test cases.

This document attempts to explain the inner workings of the THUMP program. It will be assumed that the reader is already familiar with the program as described in the above reference. The abbreviation "CA" will be used frequently to stand for "cell assembly."

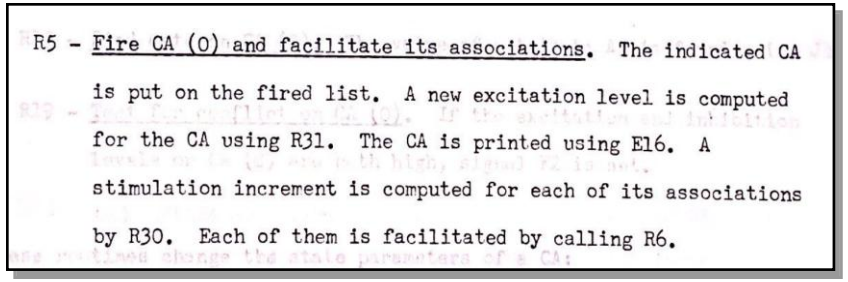


```
C1      9-1 0      9-1  9-2      9-2  9-3      9-3   0
                    A1              D1            C3
                    9-11            C3            9-31
                    A2              D3            C2
                    9-12            C2 0          9-32 0
                    A3
                    9-13
                    A4
                    9-14
                    A5
                    9-15 0

9-11 +01  0                         9-31 +01  50
9-12 +01  0                         9-32 +01  40
9-13 +01  60
9-14 +01  60
9-15 +01  0
```

Compressed: C1, A1 0 A2 0 A3 60 A4 60 A5 0, D1 C3 D3 C2, C3 50 C2 40 )

Fig. 2. A Sample Cell Assembly



R5 - Fire CA (0) and facilitate its associations. The indicated CA is put on the fired list. A new excitation level is computed for the CA using R31. The CA is printed using E16. A stimulation increment is computed for each of its associations by R30. Each of them is facilitated by calling R6.

Richard Shoup Collection, Carnegie Mellon University Library. Box 4. "ARGUS" Folder.

# A more speculative approach:

**think-aloud**

From introspection, to IP models of cognition, into HCI (Nielsen et al., 2002)

**Chain-of-Thought (CoT)**

Large-language “reasoners” improve their performance when asked to narrate their thinking (Wang et al., 2023)

# Example output

```
...
[00:03:05] Now let me just think about the form of
the piece... Do I want it to be more traditional,
with an ABA structure? Or do I want to try something
a bit more innovative?
[00:03:10] Hmm... maybe something like a theme and
variations would create some nice interest.
(thinking out loud) Yeah, that's it! That way the
Lied can really explore all the different emotions
and ideas associated with the moment
...
```

# A synthetic "think-aloud" corpus

**Prompt template**: “You are a {gender} {specialty}. You are composing a {genre}. [...] You are participating in a cognitive science experiment and you are asked to "think-aloud" while you complete this task. You are recording the thoughts that occur to you as you complete this task. [...]”

480 documents, ~ 2.8M words. Conditions:

–male, female

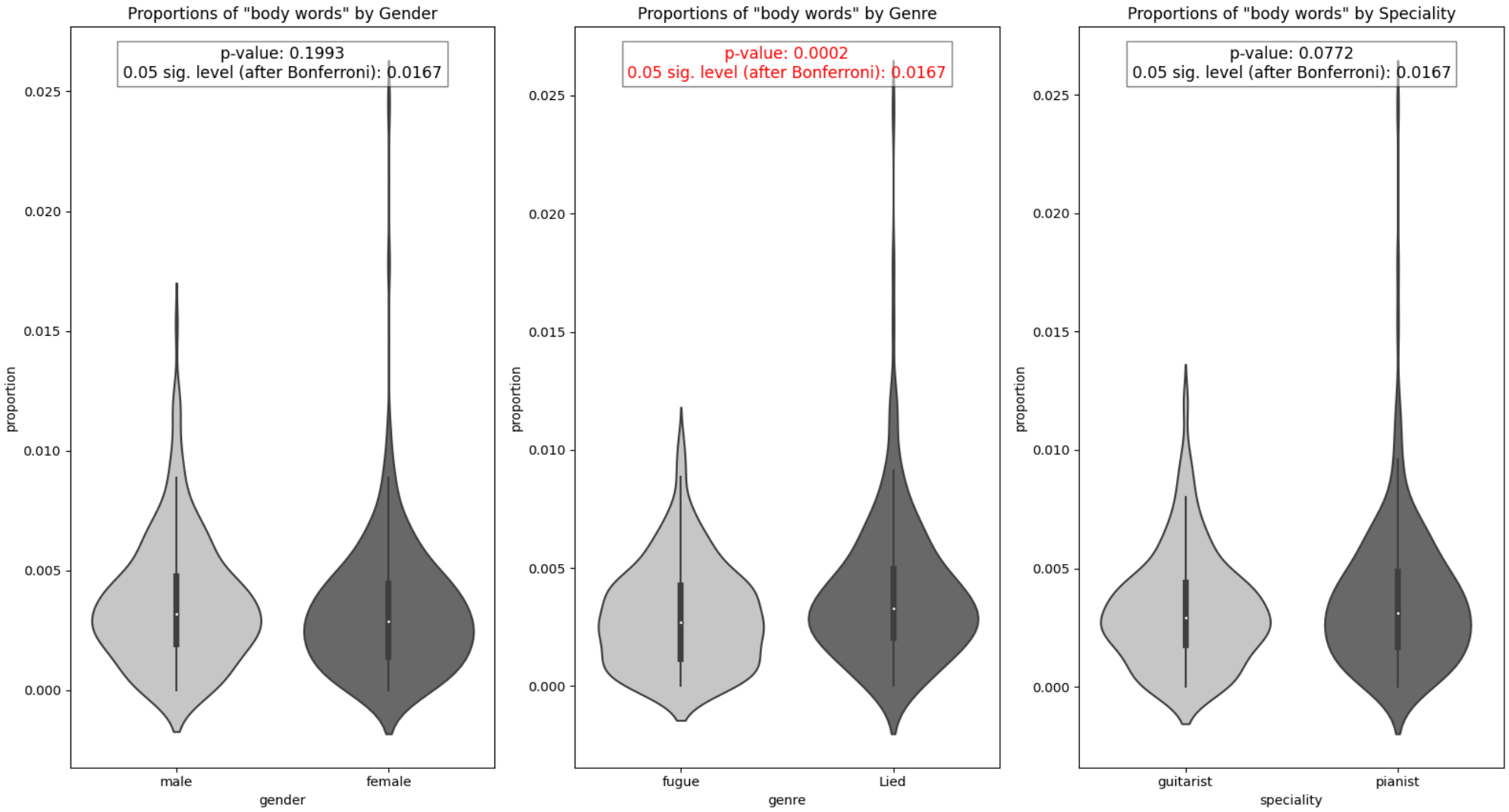
–fugue, Lied

–pianist, guitarist

Then count proportion of lemmatized “body” words and compare averages between conditions

5-, 10- and 60-minute transcripts. 20 runs on each condition; Naive “memory” to enforce consistency. Model: ollama3.1-8b-instruct-q4_0 (Dubey et al. 2024) Inference env.: ollama v. 0.3.10, default temp., RTX 3080 10GB. Run duration: ~18h.

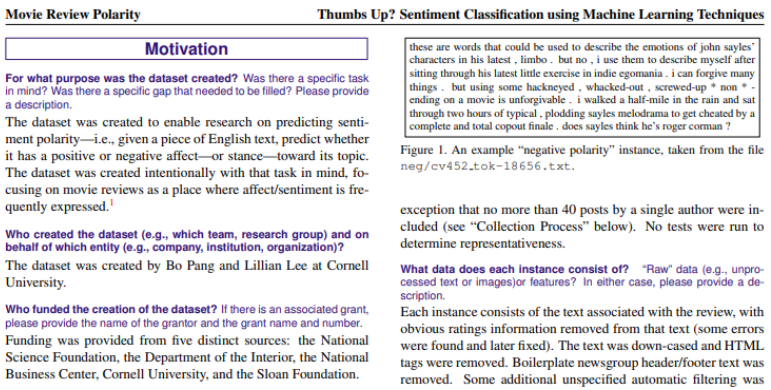
# A synthetic "think-aloud" corpus



# Wrap up

Making the past present in contemporary MIR research

# Artifacts to accompany models and datasets


Movie Review Polarity — Thumbs Up? Sentiment Classification using Machine Learning Techniques

**Motivation**

**For what purpose was the dataset created?** Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description.

The dataset was created to enable research on predicting sentiment polarity—i.e., given a piece of English text, predict whether it has a positive or negative affect—or stance—toward its topic. The dataset was created intentionally with that task in mind, focusing on movie reviews as a place where affect/sentiment is frequently expressed.[1]

**Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)?**

The dataset was created by Bo Pang and Lillian Lee at Cornell University.

**Who funded the creation of the dataset?** If there is an associated grant, please provide the name of the grantor and the grant name and number.

Funding was provided from five distinct sources: the National Science Foundation, the Department of the Interior, the National Business Center, Cornell University, and the Sloan Foundation.

these are words that could be used to describe the emotions of john sayles' characters in his latest , limbo . but no , i use them to describe myself after sitting through his latest little exercise in indie egomania . i can forgive many things . but using some hackneyed , whacked-out , screwed-up * non * - ending on a movie is unforgivable . i walked a half-mile in the rain and sat through two hours of typical , plodding sayles melodrama to get cheated by a complete and total copout finale . does sayles think he's roger corman ?

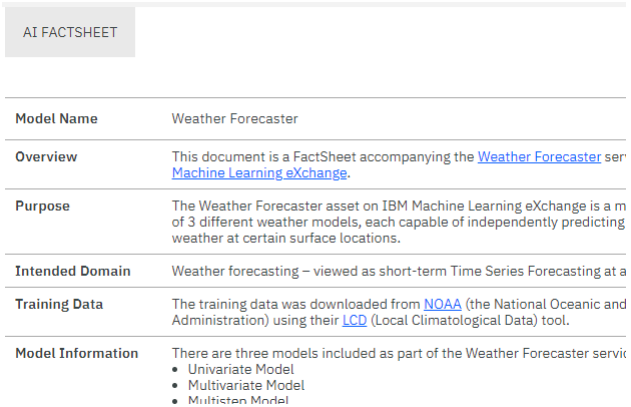
Figure 1. An example "negative polarity" instance, taken from the file `neg/cv452_tok-18656.txt`.

exception that no more than 40 posts by a single author were included (see "Collection Process" below). No tests were run to determine representativeness.

**What data does each instance consist of?** "Raw" data (e.g., unprocessed text or images)or features? In either case, please provide a description.

Each instance consists of the text associated with the review, with obvious ratings information removed from that text (some errors were found and later fixed). The text was down-cased and HTML tags were removed. Boilerplate newsgroup header/footer text was removed. Some additional unspecified automatic filtering was

(Hutchinson et al. 2021)


AI FACTSHEET

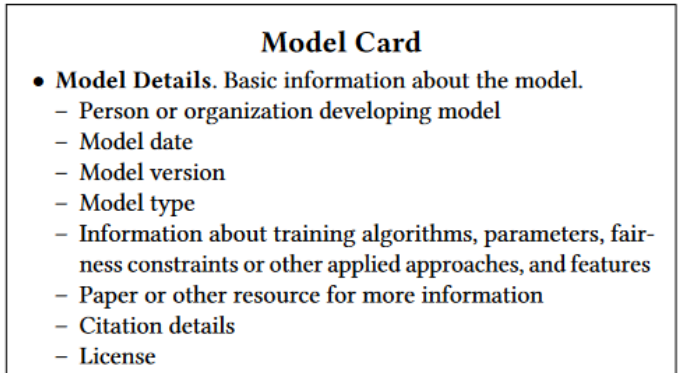
| | |
|---|---|
| Model Name | Weather Forecaster |
| Overview | This document is a FactSheet accompanying the Weather Forecaster ser Machine Learning eXchange. |
| Purpose | The Weather Forecaster asset on IBM Machine Learning eXchange is a m of 3 different weather models, each capable of independently predicting weather at certain surface locations. |
| Intended Domain | Weather forecasting – viewed as short-term Time Series Forecasting at a |
| Training Data | The training data was downloaded from NOAA (the National Oceanic and Administration) using their LCD (Local Climatological Data) tool. |
| Model Information | There are three models included as part of the Weather Forecaster servic<br>• Univariate Model<br>• Multivariate Model<br>• Multistep Model |

Datasheets for Datasets (Gebru et al. 2020)

↪ **Artsheets for Art Datasets (Srinivasan et al. 2021)**

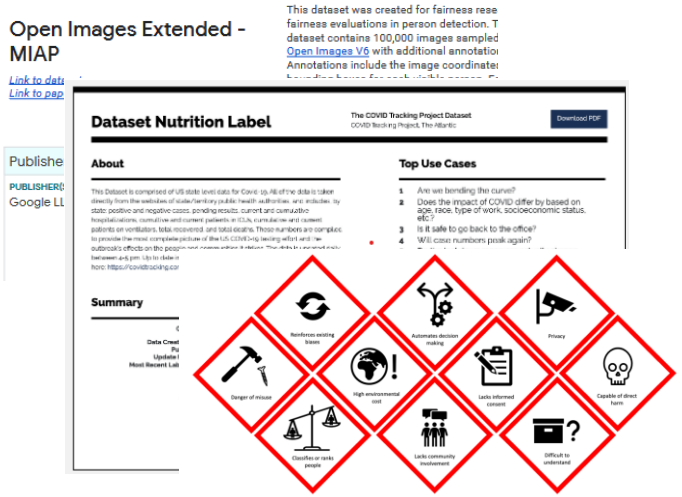
↪ Healthsheet (Rostamzadeh et al. 2022)

[IBM] AI Factsheet (Arnold et. al 2019)


**Model Card**

- **Model Details.** Basic information about the model.
  - Person or organization developing model
  - Model date
  - Model version
  - Model type
  - Information about training algorithms, parameters, fairness constraints or other applied approaches, and features
  - Paper or other resource for more information
  - Citation details
  - License

Model Cards for Model Reporting (Mitchell et al. 2019)




Data Cards (Pushkarna and Zaldivar 2021)

Dataset Nutrition Labels (Chmielinski et al. 2020)

Data Hazard Labels (Natalie Thurlby and Nina Di Cara)

# Requirements for a new artifact?

- Make **existing** histories available at the point of research design
  - …not as an afterthought,
  - …nor just as part of literature review

**OR**

- Make tools to support the creation of **new** history where no relevant candidate exists

**AND**

- Make **history** consumable in the context of a finished scientific paper/model/dataset release

Is this possible?

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**AND**

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Is this possible?

…desirable?

# References (1)


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